

SWE308: Information Security
Lecture 9: Asymmetric Cryptography
Encrypt your message with a LOCK
and
Decrypt the message with the corresponding KEY

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Presentation Outline

- 1 Symmetric Cryptography: How to Establish the Key?
- 2 Key Agreement over an Insecure Channel
- 3 Asymmetric Ciphers
- 4 Can We Use A Combination of Symmetric and Asymmetric Cryptosystem?
- 5 Summary

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Symmetric Ciphers

Physical security metaphor

- Shared key = combination to a lock
- Storage:
 - ① Put letter into the box, and
 - ② Lock box with combination
 - ③ Later, retrieval needs combination
- Transmission:
 - ① Alice: Put letter in box, lock with combination. Send the locked box to Bob
 - ② Opens box with same combination Alice used to lock it. Takes letter out of box and reads it



Symmetric Ciphers and Security

Security services provided by ciphers depend on the security of the key - **Cryptographic techniques provide no security if the key is compromised!**

The security of the symmetric key involves:

- **Requires confidentiality?**
 - Who should have access to the key?
- **Requires integrity assurance?**
 - What happens if the key is altered?
 - What happens if an attacker replaces a key with a key that they know?
- **Requires availability?**
 - What happens if an attacker destroys the key?

Symmetric Ciphers and Security

If we use symmetric ciphers to provide security services for information, then we must also provide security services for the symmetric keys used with the ciphers.

- Effectively, replaced one security problem with another.

Cryptographic techniques can be used to provide security for cryptographic keys.

- Store keys in a file which we encrypt (using another key) to provide confidentiality
 - Forms a hierarchy of keys: master keys, session keys, etc
 - Like keeping your house key, car key, etc in a locked box
- Store hash values or MACs of secret key files and use to detect modifications to the key files, etc.

The Key Establishment Problem

Scenario:

- Alice and Bob want to communicate securely.
- They have to send messages over an insecure channel.
 - MITM Eve may be eavesdropping, or modifying transmissions.
- They decide to use cryptographic mechanisms based on symmetric ciphers to provide security services for their communications.
- Before they can have a secure communication session, they both need to have a shared secret key: a session key.

Question: How can they do this securely, if the communication channel they are using is not secure?

The Key Establishment Problem

Option 1 - Secure pre-distribution

- Distribute secret key 'out-of-band':
 - Not over the insecure communication channel, but over a different secure channel.
 - Maybe physical distribution via trusted courier?

How many secret keys are needed in a system?

- 2 people, only 1 secret key is needed.
- 3 people, and each participant may possibly communicate securely with every other participant, 3 secret keys are needed.
- 4 people, 6 secret keys are needed.
- n people, $\frac{n(n-1)}{2}$ secret keys are needed

The Key Establishment Problem

Option 1 - Secure pre-distribution

- n people, $\frac{n(n-1)}{2}$ secret keys are needed
- What sort of numbers are we looking at?
 - for $n = 5$, $(5 \times 4)/2 = 10$ keys must be established.
 - for $n = 10$, $(10 \times 9)/2 = 45$ keys must be established.
 - for $n = 20$, $(20 \times 19)/2 = 190$ keys must be established.
 - for $n = 100$, $(100 \times 99)/2 = 4\,950$ keys.
 - for $n = 200$, $(200 \times 199)/2 = 19\,900$ keys.
 - for $n = 350$, $(350 \times 349)/2 = 61\,075$ keys.
- **Not practical:**
 - for large numbers of participants, or
 - if users will be added/removed from system frequently

The Key Establishment Problem

Option 2 - Trusted third party

- Alternative to everyone having a secret key for possible communications with everyone else:
 - Have one trusted party
 - Everyone holds shared secret keys to communicate with that party
 - If you want to communicate securely with someone else:
 - You ask the trusted third party (TTP) to send you a key (encrypted using the secret key you share with the third party)
 - Then you decrypt the TTP message to obtain that new key (the other person you want to communicate with needs it too),
 - Now you can use this key in communications with the other party
- Kerberos is a well known scheme that takes this approach

The Key Establishment Problem

Option 3 - Diffie-Hellman key agreement algorithm

- 1976: Whitfield Diffie & Martin Hellman published a radical method for forming symmetric crypto keys
- Made use of certain mathematical methods:
 - modular exponentiation, with careful parameter choices
- Two parties:
 - without prior agreement, and
 - without a secure distribution channel
 - can form a shared secret only they know
- To do this, each party:
 - 1 Chooses their own secret value
 - 2 Sends the other a mathematical function of their chosen value,
 - 3 Combines their own value with the received information to form the shared secret value
- An eavesdropper **cannot** determine the shared secret by listening to the messages exchanged by the two parties
 - unless they can break the discrete log problem

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Secure Communications over Insecure Channel - Merkle's Puzzle

Programming
Techniques

S. L. Graham, R. L. Rivest
Editors

Secure Communications Over Insecure Channels

Ralph C. Merkle
Department of Electrical Engineering and
Computer Sciences
University of California, Berkeley

According to traditional conceptions of cryptographic security, it is necessary to transmit a key, by secret means, before encrypted messages can be sent securely. This paper shows that it is possible to select a key over open communications channels in such a fashion that communications security can be maintained. A method is described which forces any enemy to expend an amount of work which increases as the square of the work required of the two communicants to select the key. The method provides a logically new kind of protection against the passive eavesdropper. It suggests that further research on this topic will be highly rewarding, both in a theoretical and a practical sense.

Key Words and Phrases: security, cryptography, cryptology, communications security, wiretap, computer network security, passive eavesdropping, key distribution, public key cryptosystem

CR Categories: 3.56, 3.81

An early construction for a public-key cryptosystem was conceived, a protocol devised by Ralph Merkle in 1974 and published in 1978

R. Merkle, "Secure communication over an insecure channel", Communications of the ACM, April 1978.

<http://www.merkle.com/1974/PuzzlesAsPublished.pdf>

Diffie-Hellman Key Agreement

Diffie and Hellman published “New Directions in Cryptography” Nov '76: “We stand today at the brink of a revolution in cryptography.”

- “Public-Key Cryptosystem” This remarkable idea developed jointly with Merkle.

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IEEE TRANSACTIONS ON INFORMATION THEORY, VOL. IT-22, NO. 6, NOVEMBER 1976

New Directions in Cryptography

Invited Paper

WHITFIELD DIFFIE AND MARTIN E. HELLMAN, MEMBER, IEEE



Abstract—Two kinds of contemporary developments in cryptography are examined. Widening applications of teleprocessing have given rise to a need for new types of cryptographic systems, which minimize the need for secure key distribution channels and supply the equivalent of a written signature. This paper suggests ways to solve these currently open problems. It also discusses how the theories of communication and computation are beginning to provide the tools to solve cryptographic problems of long standing.

I. INTRODUCTION

WE STAND TODAY on the brink of a revolution in cryptography. The development of cheap digital

The best known cryptographic problem is that of privacy: preventing the unauthorized extraction of information from communications over an insecure channel. In order to use cryptography to insure privacy, however, it is currently necessary for the communicating parties to share a key which is known to no one else. This is done by sending the key in advance over some secure channel such as private courier or registered mail. A private conversation between two people with no prior acquaintance is a common occurrence in business, however, and it is unrealistic to expect initial business contacts to be postponed long enough for keys to be transmitted by some physical means. The cost and delay imposed by this key distribution problem is a major barrier to the transfer of business

<https://ee.stanford.edu/~hellman/publications/24.pdf>

Diffie-Hellman Key Agreement

for system parameters p (a prime) and g

Alice picks random integer a



Alice computes shared secret
 $(g^b)^a = g^{ab} \bmod p$

Bob picks random integer b



Bob computes same secret
 $(g^a)^b = g^{ab} \bmod p$

But eavesdropper Eve can't compute g^{ab}
 even if she knows g^a and g^b

Diffie-Hellman Key Agreement

1. Alice and Bob have common system parameters p and g .
2. Alice and Bob each generate random secret exponents a and b .
3. Alice and Bob each compute public messages $g^a \bmod p$ and $g^b \bmod p$.



Common paint

+

Secret colour

=

Alice's public colour



Common paint

+

Secret colour

=

Bob's public colour

Diffie-Hellman Key Agreement

1. Alice and Bob have common system parameters p and g .
2. Alice and Bob each generate random secret exponents a and b .
3. Alice and Bob each compute public messages $g^a \bmod p$ and $g^b \bmod p$.
4. Alice and Bob exchange public messages.
5. Alice and Bob use their secret exponent and the other person's public message to compute the shared secret.



Secret colour

+

Bob's public colour

=

Shared secret colour

Secret colour

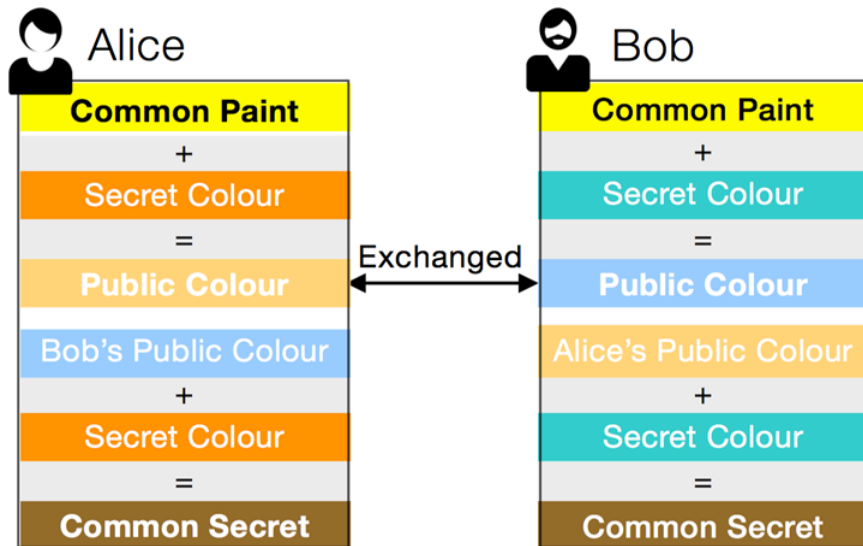
+

Alice's public colour

=

Shared secret colour

Diffie-Hellman Key Agreement



Diffie-Hellman Key Agreement: Example

Non-secret values in blue, and secret values in red. ¹

- g = public (prime) base, known to Alice, Bob, and Eve. $g = 5$
- p = public (prime) modulus, known to Alice, Bob, and Eve. $p = 23$
- a = Alice's private key, known only to Alice. $a = 6$
- b = Bob's private key known only to Bob. $b = 15$
- A = Alice's public key, known to Alice, Bob, and Eve. $A = g^a \bmod p = 8$
- B = Bob's public key, known to Alice, Bob, and Eve. $B = g^b \bmod p = 19$

Alice		Bob		Eve	
Known	Unknown	Known	Unknown	Known	Unknown
$p = 23$		$p = 23$		$p = 23$	
$g = 5$		$g = 5$		$g = 5$	
$a = 6$	b	$b = 15$	a		a, b
$A = 5^a \bmod 23$		$B = 5^b \bmod 23$			
$A = 5^6 \bmod 23 = 8$		$B = 5^{15} \bmod 23 = 19$			
$B = 19$		$A = 8$		$A = 8, B = 19$	
$s = B^a \bmod 23$		$s = A^b \bmod 23$			
$s = 19^6 \bmod 23 = 2$		$s = 8^{15} \bmod 23 = 2$			s

¹Source: https://en.wikipedia.org/wiki/Diffie%E2%80%93Hellman_key_exchange

Diffie-Hellman Key Agreement

Option 3 - Diffie-Hellman key agreement algorithm

- Q: Does that mean Alice and Bob can now:
 - Establish a shared secret key over an insecure channel?

Diffie-Hellman Key Agreement

Option 3 - Diffie-Hellman key agreement algorithm

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 - Establish a shared secret key over an insecure channel?
 - YES!

Diffie-Hellman Key Agreement

Option 3 - Diffie-Hellman key agreement algorithm

- Q: Does that mean Alice and Bob can now:
 - Establish a shared secret key over an insecure channel?
 - YES!
- And communicate securely, without worrying about Eve (the potential MITM attacker)?

Diffie-Hellman Key Agreement

Option 3 - Diffie-Hellman key agreement algorithm

- Q: Does that mean Alice and Bob can now:
 - Establish a shared secret key over an insecure channel?
 - YES!
- And communicate securely, without worrying about Eve (the potential MITM attacker)?
 - No

Diffie-Hellman Key Agreement

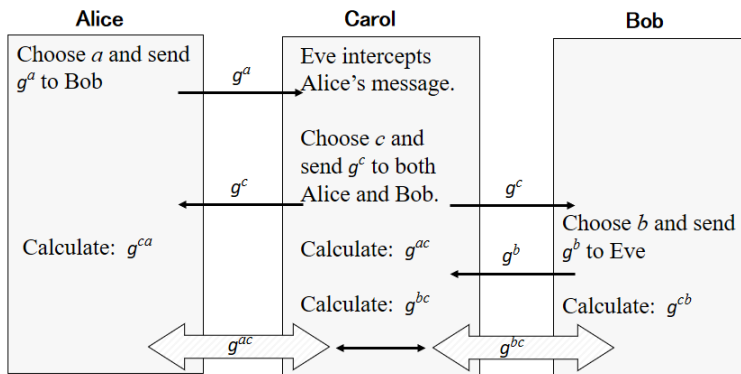
Option 3 - Diffie-Hellman key agreement algorithm

- Q: Does that mean Alice and Bob can now:
 - Establish a shared secret key over an insecure channel?
 - YES!
- And communicate securely, without worrying about Eve (the potential MITM attacker)?
 - No
- If Eve is clever, she can still eavesdrop ...
 - Problem is: no authentication of the communicating parties
 - While they communicate to establish the key, Alice and Bob have no assurance about who they are communicating with (Alice does not know for sure it is Bob, and vice versa)
 - So a man-in-the-middle (Eve) attack is possible

Diffie-Hellman Key Agreement

Diffie-Hellman key agreement & MITM attack

- Eve completes two separate Diffie-Hellman key agreements, and now shares keys with both Alice and Bob
- Alice sends confidential messages believing that she is communicating with Bob. Eve can decrypt these messages, read them and re-encrypt the message for Bob and send it on
- Likewise, Bob sends messages thinking he is communicating with Alice, and is unaware of Eve



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Where did the asymmetric idea come from?

Diffie Hellman key establishment:

- Used for two parties to establish a shared secret
- But vulnerable to MITM

Important aspect of Diffie-Hellman is application of a new type of mathematics in cryptography

- Idea that (for appropriate functions) could have public components without compromising confidentiality
- Lead to asymmetric cipher systems: New type of crypto and new functionality

Asymmetric Ciphers

Alice has a lock, a key that opens the lock, and a box! What can Alice do to ensure that Bob (or anybody else) can send her a postal mail such that the postman won't be able to read the mail?



Image source: <https://brilliant.org/wiki/public-key-cryptography/>

Asymmetric Ciphers

Symmetric ciphers:

- encryption & decryption keys are the same

Asymmetric ciphers:

- encryption & decryption keys are different:
 - Keys are related, but it is **computationally infeasible** to derive one key from the other
- Each participant needs a pair of keys:
 - One key kept private: K_{Apriv} & other key is made public: K_{Apub}
 - For example, Alice has (K_{Apriv}, K_{Apub})
 - Asymmetric crypto also known as public key cryptography
- Security of asymmetric cryptosystem depends on:
 - The strength of the algorithm,
 - The key size, and
 - Confidentiality of the user's private key K_{Apriv} (or K_{Bpriv} or K_{Cpriv} ...)

Asymmetric Cryptography

Key distribution problem solved!

- Key distribution is much simpler than for symmetric ciphers
 - as anyone may know the public key
- Each participant needs to:
 - create their asymmetric key pair
 - publish the public key
 - keep the private key secret
- If there are n participants, then a total of n key pairs must be created

Using Asymmetric Ciphers

How do you get to know someone's public key?

- Give it to you directly (have it in their email signature)
- Put it up on their website
- Make it available through a public keyserver
 - Example: <http://pgp.mit.edu:11371/>

We will discuss this more when we do the lecture on public key infrastructure (PKI).

Using Asymmetric Ciphers for Confidentiality

For Alice to send a confidential message to Bob, Alice needs to:

- 1 Know Bob's public key K_{Bpub}
- 2 Encrypt the plaintext message P using the asymmetric encryption algorithm and Bob's public key K_{Bpub}
- 3 Send the resulting ciphertext to Bob

For Bob to recover the message,

- 1 Bob uses his private key K_{Bpriv} and the asymmetric decryption algorithm to decrypt the ciphertext

Who can encrypt the message and send to Bob? Who can decrypt the message?

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Who can encrypt the message and send to Bob? Who can decrypt the message?

- **Anyone** who knows Bob's public key can encrypt a message and send to Bob
- **Only Bob** knows the corresponding private key, so **only Bob** can decrypt the encrypted message

Using Asymmetric Ciphers for Confidentiality

Metaphor:

- Bob's public key = padlock, which he gives to Alice
- Bob's private key = the key to the padlock, which he keeps

Encryption:

- Alice puts a message in a box and locks it with Bob's padlock
- Only Bob has the key to open the padlock

1. Bob creates public key / private key pair.



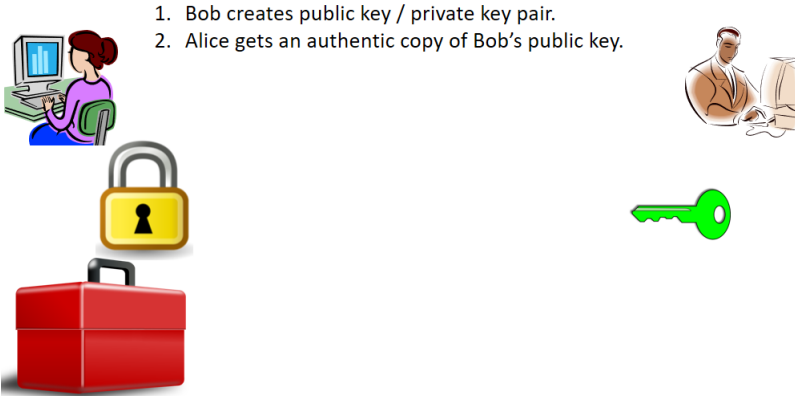
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2. Alice gets an authentic copy of Bob's public key.
3. Alice encrypts her message using Bob's public key.



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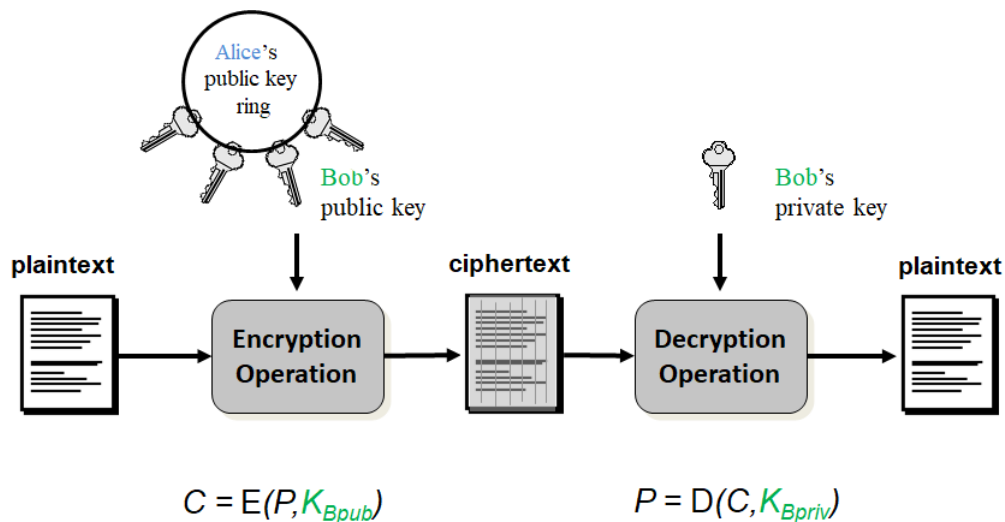


1. Bob creates public key / private key pair.
2. Alice gets an authentic copy of Bob's public key.
3. Alice encrypts her message using Bob's public key.
4. Alice sends the ciphertext to Bob.
5. Bob decrypts using his private key.



Asymmetric ciphers for confidentiality

For a message Alice sends to Bob



Asymmetric ciphers for confidentiality

Notation

- Encryption: $C = E(P, K_{Rpub})$
- Decryption: $P = D(C, K_{Rpriv})$

Encryption and Decryption Keys:

- Public key of the recipient K_{Rpub} is used for encryption
- Private key of the recipient K_{Rpriv} is used for decryption

The private key must remain confidential (i.e. known only to its owner)

- Anyone is allowed to know the public key
- Only the owner may know the associated private key

Asymmetric Ciphers: Example

Ron Rivest, Adi Shamir, and Leonard Adleman published "A Method for Obtaining Digital Signatures and Public-Key Cryptosystems" in 1977

Programming Techniques
S.L. Graham, R.L. Rivest*
Editors

A Method for Obtaining Digital Signatures and Public-Key Cryptosystems

R. L. Rivest, A. Shamir, and L. Adleman
MIT Laboratory for Computer Science
and Department of Mathematics

An encryption method is presented with the novel property that publicly revealing an encryption key does not thereby reveal the corresponding decryption key. This has two important consequences:

(1) Couriers or other secure means are not needed to transmit keys, since a message can be enciphered using an encryption key publicly revealed by the intended recipient. Only he can decipher the message, since only he knows the corresponding decryption key. (2) A message can be "signed" using a privately held decryption key. Anyone can verify this signature using the corresponding publicly revealed encryption key. Signatures cannot be forged, and a signer cannot later deny the validity of his signature. This has obvious applications in "electronic mail" and "electronic funds transfer" systems. A message is encrypted by representing it as a number M , raising M to a publicly specified power e , and then taking the remainder when the result is divided by the publicly specified product, n , of two large secret prime numbers p and q . Decryption is similar; only a different, secret, power d is used, where $e \cdot d = 1 \pmod{(p-1) \cdot (q-1)}$. The security of the system rests in part on the difficulty of factoring the published divisor, n .

Key Words and Phrases: digital signatures, public-key cryptosystems, privacy, authentication, security, factorization, prime number, electronic mail, message-passing, electronic funds transfer, cryptography.

CR Categories: 2.12, 3.15, 3.50, 3.81, 5.25

RSA Cryptosystem

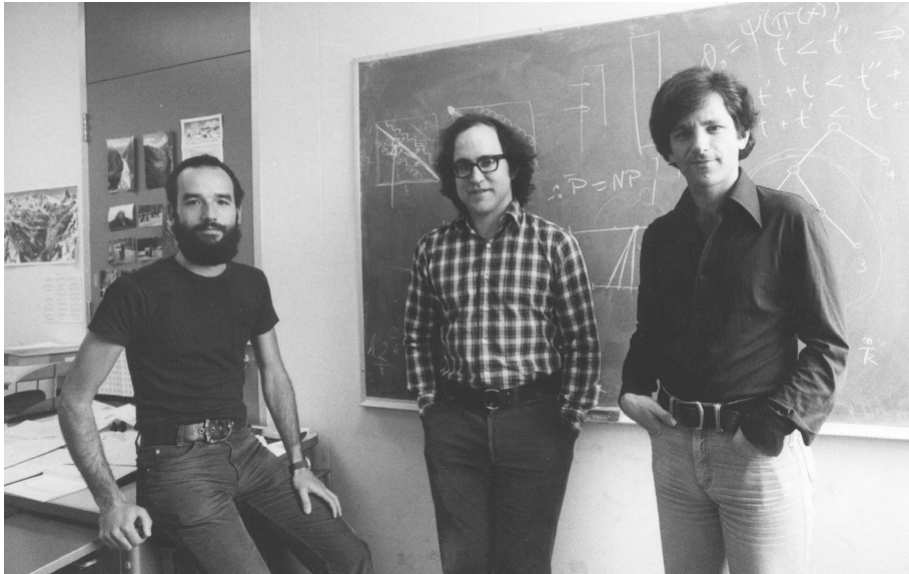
- Rivest-Shamir-Adleman, MIT, 1977
- Most well-known asymmetric cryptosystem
- Can be used for
 - encryption (for confidentiality)
 - digital signature scheme (for message integrity, authenticating sender)
- Based on difficulty of factorising large integers
 - Calculations performed using modular arithmetic
 - Where the modulus is the product of two large primes

Watch:

<https://www.youtube.com/watch?v=mv0sb9vNIWM>

Rivest, R., Shamir, A., and Adleman, L. A method for obtaining digital signatures and public-key cryptosystems. Comm. A CM 21, 2 (Feb. 1978), 120-126.

Invention of RSA



<https://people.csail.mit.edu/rivest/pubs/ARS03.rivest-slides.pdf>

"Textbook" RSA Cryptosystem

Key Generation

Select p, q	p and q both prime, $p \neq q$
Calculate $n = p \times q$	
Calculate $\phi(n) = (p - 1)(q - 1)$	
Select integer e	$\gcd(\phi(n), e) = 1; 1 < e < \phi(n)$
Calculate d	$de \bmod \phi(n) = 1$
Public key	$KU = \{e, n\}$
Private key	$KR = \{d, n\}$

Encryption

Plaintext:	$M < n$
Ciphertext:	$C = M^e \pmod{n}$

Decryption

Ciphertext:	C
Plaintext:	$M = C^d \pmod{n}$

RSA visualization: <https://www.cryptool.org/en/cto/rsa-visual>

RSA step-by-step: <https://www.cryptool.org/en/cto/rsa-step-by-step>

"Textbook" RSA Cryptosystem: Example

Consider the following example:

- ① Let, the two primes be: $p = 3$ and $q = 11$
- ② Compute the modulus: $n = p \times q = 3 \times 11 = 33$
- ③ Compute Euler's totient function: counts the positive integers up to a given integer n that are relatively prime to n . In this example $n = 33$, so we have:
 - 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33
 - There is an easy way to compute: $\phi(n) = (p - 1)(q - 1) = (3 - 1)(11 - 1) = 20$
 - So, 20 positive integers up to 33 that are relatively prime to 33
- ④ Let, the public exponent $e = 7$
- ⑤ Determine the private exponent d such that:
 - $de \bmod \phi(n) = 1$
 - $7d \bmod \phi(n) = 1$
 - $d = 3$

Let's encrypt an integer $M = 8$

- $C = M^e \bmod n = 8^7 \bmod 33 = 2$

Let's now decrypt the ciphertext C

- $M = C^d \bmod n = 2^3 \bmod 33 = 8$

Security of Textbook RSA?

Security of Textbook RSA

- This scheme is deterministic—cannot be CPA-secure!
- c is not uniform unless m is—easy to compute eth root of $c = [m^e \bmod N]$ when m is small
- Partial information about the message may be learned by attackers

Textbook RSA should never be used!

How to fix?²

- Use padding schemes such as OAEP (Optimal Asymmetric Encryption Padding) before encryption
- Add randomness to the plaintext, making RSA encryption probabilistic

²See: <https://datatracker.ietf.org/doc/html/rfc8017>

RSA: The Factoring Problem

RSA security indirectly relies on the fact that prime factorization is computationally very hard

- The reason factorization remains a challenging problem is the size of numbers that are used in RSA
- Currently (February 2020) the largest number that has been factored is 829 bits (250 decimal digits)
- RSA keys are generally at least 1024 bits long (309 decimal digits)
- Since 2015, NIST recommends a minimum key size of 2048 bits for RSA

Theoretically Shor's algorithm can solve it in polynomial time, but only works with a Quantum computer!

https://en.wikipedia.org/wiki/RSA_Factoring_Challenge

Symmetric and Asymmetric Ciphers: Key Length Comparison

Table: Symmetric and asymmetric ciphers offering comparable security

AES Key Size	RSA Key Size
-	1024
128	3072
192	7680
256	15360

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Confidentiality Using A Hybrid Cryptosystem

Symmetric ciphers are much faster than currently available asymmetric ciphers

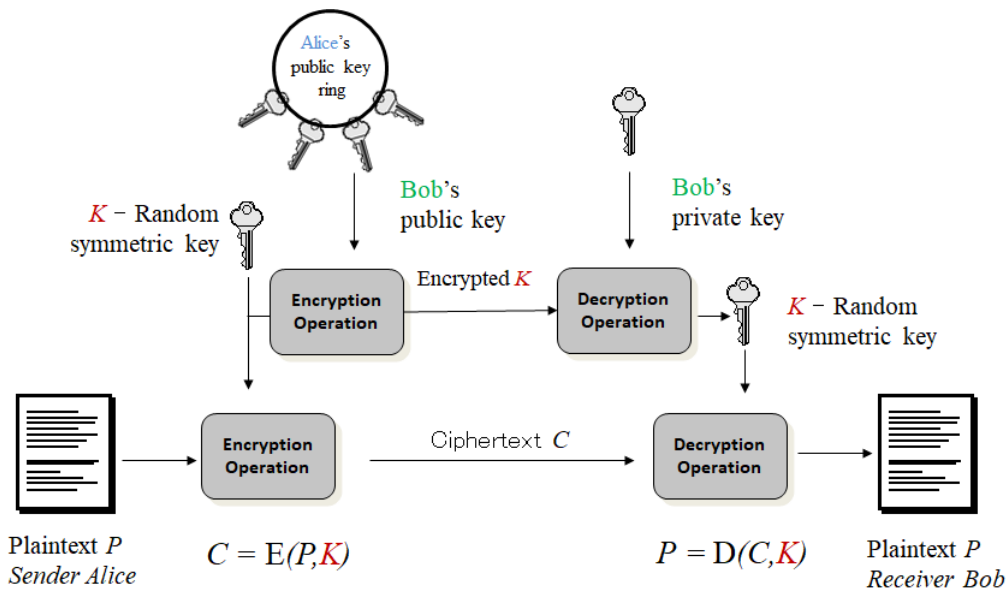
- because they are less computationally expensive

However, asymmetric ciphers greatly simplify key distribution

So, many cryptosystems use a combination: Hybrid cryptosystems

- First, the asymmetric cipher is used to provide confidentiality for a particular short message:
 - a randomly chosen shared secret key to be used with a particular symmetric cipher
- Then the symmetric cipher is used with that shared secret key for encrypting the bulk data

Confidentiality Using A Hybrid Cryptosystem



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End of Lecture 9! 😊

Summary

- Diffie-Hellman key agreement algorithm proposed as a solution to the symmetric key distribution problem
 - Based on modular exponentiation
- New thinking about mathematical functions lead to new type of cryptosystem: asymmetric cryptography
 - Two keys, one to be made public, one to be kept private (hence the name: public key cryptography)
 - Key distribution problem solved
- Asymmetric crypto is comparatively slower than the symmetric crypto
 - A hybrid cryptosystem can be used by taking advantage of both

Remember to complete your quiz this week!

Next Week Lecture

- Asymmetric crypto to provide integrity assurance and user authentication
- Public Key Infrastructure (PKI)

QUESTIONS?

Appendix: University's vision and mission

- ① Vision: Xiamen University Malaysia aspires to become a university with a distinct global outlook, featuring first-class teaching and research, and embracing cultural diversity.
- ② Mission: To nurture young talents with dignity and wisdom, turning them into fine citizens of the region who will contribute to the prosperity of the people and social progress of Malaysia, China and Southeast Asia.